

SE LAB - 2

Vihaan Wadhwa

PES1UG24AM319

Sec-F

Problem Statement - 19

Personalized Meal & Diet Subscription Manager

Agile Backlog — Epics:

EPIC-1: Dietary Profile Management

Enable subscribers to create and maintain their dietary profile — including allergies, diabetic limits, calorie goals, and macro-nutrient constraints — so the system can generate meal plans tailored to their health needs.

Lab 1 mapping: FR-002

EPIC-2: Meal Plan Generation & Validation

Automatically generate weekly meal plans that strictly comply with a subscriber's macro-nutrient constraints and allergen exclusions, and allow dietitians to review a subscriber's constraints alongside their generated plan.

Lab 1 mapping: FR-001, FR-005

EPIC-3: Delivery Scheduling & Tracking

Allow subscribers to pause, skip, or reschedule upcoming meal deliveries up to 12 hours before dispatch, and track and display the fulfillment status of each scheduled delivery.

Lab 1 mapping: FR-003, FR-004, NFR-001

EPIC-4: Security & Access Control

Restrict access to subscriber dietary profiles and meal-plan information to authenticated and authorized users only, protecting sensitive health data.

Lab 1 mapping: NFR-002

User Stories, Priorities & Story Points:

ID	Title	User Story	Priority	SP	Req.
1.1	Create Dietary Profile	As a subscriber, I want to enter my allergies, diabetic limits, calorie goals, and macro-nutrient constraints, so that the system can generate meal plans tailored to my dietary needs.	High	5	FR-002
1.2	Update Dietary Profile	As a subscriber, I want to update my dietary profile at any time, so that my meal plans stay accurate as my needs change.	Medium	3	FR-002
2.1	Generate Weekly Meal Plan	As a subscriber, I want the system to generate a weekly meal plan based on my dietary constraints, so that I receive meals that fit my health requirements without manual planning.	High	8	FR-001
2.2	Validate Meal Plan Against Constraints	As a subscriber, I want the system to check that my generated meal plan excludes my allergens and meets my macro-nutrient limits, so that I can trust the plan is safe to follow.	High	5	FR-001
2.3	Review Subscriber's Meal Plan (Dietitian)	As a dietitian, I want to view a subscriber's dietary constraints and their generated weekly meal plan, so that I can verify it's appropriate for their health needs.	Medium	3	FR-005
3.1	Pause, Skip, or Reschedule Delivery	As a subscriber, I want to pause, skip, or reschedule an upcoming meal delivery, so that I can manage my subscription around my schedule.	High	5	FR-003
3.2	Modify Delivery Within Cutoff Window	As a subscriber, I want to make schedule changes up to 12 hours before dispatch, so that I have a fair window to adjust my deliveries.	High	3	NFR-001
3.3	Track Delivery Fulfillment Status	As a subscriber, I want to see the fulfillment status of each scheduled delivery, so that I know whether my meal has been delivered.	High	3	FR-004
4.1	Restrict Access to Dietary Data	As a subscriber, I want my dietary and meal-plan information to be visible only to authenticated and authorized users, so that my sensitive health data stays private.	High	5	NFR-002
4.2	Authenticated Access for Dietitians	As a dietitian, I want to log in securely before accessing any subscriber's data, so that patient information is protected from unauthorized access.	High	3	NFR-002

Sprint Plan:

Sprint 1: 24 points — dietary profile setup and meal plan generation

- ST-1.1 — Create Dietary Profile (5 SP)
- ST-1.2 — Update Dietary Profile (3 SP)
- ST-2.1 — Generate Weekly Meal Plan (8 SP)
- ST-2.2 — Validate Meal Plan Against Constraints (5 SP)
- ST-2.3 — Review Subscriber's Meal Plan — Dietitian (3 SP)

Sprint 2: 19 points — delivery management and access security

- ST-3.1 — Pause, Skip, or Reschedule Delivery (5 SP)
- ST-3.2 — Modify Delivery Within Cutoff Window (3 SP)
- ST-3.3 — Track Delivery Fulfillment Status (3 SP)
- ST-4.1 — Restrict Access to Dietary Data (5 SP)
- ST-4.2 — Authenticated Access for Dietitians (3 SP)

Total: 43 points across both sprints. Sprint 1 gets the core value delivered : a subscriber can set up their profile and receive a validated meal plan. Sprint 2 rounds it out with delivery flexibility and the security layer protecting that health data.

Jira Screenshots:

Spaces
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Summary Backlog **Board** Development Timeline Docs Forms +

Search board Filter Group Complete sprint

To Do (5)	In Progress (0)	In Review (0)	Done (0)
Story 1.1: Create Dietary Profile Parent: Epic 1: Dietary Profile Management... SCRUM-11 (5)	+ Create	+ Create	+ Create
Story 1.2: Update Dietary Profile Parent: Epic 1: Dietary Profile Management... SCRUM-12 (3)			
Story 2.1: Generate Weekly Meal Plan Parent: Epic 2: Meal Plan Generation & Val... SCRUM-13 (8)			
Story 2.2: Validate Meal Plan Against Constraints + Create			

Spaces
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Summary Backlog **Board** Development Timeline Docs Forms +

Search board Filter Group Complete sprint

To Do (0)	In Progress (5)	In Review (0)	Done (0)
+ Create	SCRUM-11 (5)	+ Create	+ Create
	Story 1.2: Update Dietary Profile Parent: Epic 1: Dietary Profile Management... SCRUM-12 (3)		
	Story 2.1: Generate Weekly Meal Plan Parent: Epic 2: Meal Plan Generation & Val... SCRUM-13 (8)		
	Story 2.2: Validate Meal Plan Against Constraints Parent: Epic 2: Meal Plan Generation & Val... SCRUM-14 (5)		
	+ Create		

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Complete sprint

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To Do0

+ Create

In Progress0

+ Create

In Review4

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...

Done0

+ Create

Story 1.1: Create Dietary Profile

Parent

Epic 1: Dietary Profile Management...

SCRUM-11

5

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Story 1.2: Update Dietary Profile

Parent

Epic 1: Dietary Profile Management...

SCRUM-12

3

⬆️

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Story 2.1: Generate Weekly Meal Plan

Parent

Epic 2: Meal Plan Generation & Val...

SCRUM-13

8

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Story 2.3: Review Subscriber's Meal Plan (Dietitian)

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Complete sprint

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To Do0

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In Progress0

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Done4

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Story 1.1: Create Dietary Profile

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Parent

Epic 1: Dietary Profile Management...

SCRUM-11

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Story 1.2: Update Dietary Profile

Parent

Epic 1: Dietary Profile Management...

SCRUM-12

3

✓

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Story 2.1: Generate Weekly Meal Plan

Parent

Epic 2: Meal Plan Generation & Val...

SCRUM-13

8

✓

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Story 2.3: Review Subscriber's Meal Plan (Dietitian)

+ Create



Complete SCRUM Sprint 1

This sprint contains 5 completed work items.
That's all of them - well done!

Cancel

Complete sprint

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Summary Backlog Board Development Timeline Docs Forms

Search backlog Filter

1 of 1 work item visible Estimate: 5 of 5

No epic

Epic 1: Dietary Profile Management (FR-002)

Epic 2: Meal Plan Generation & Validation (FR-001, FR-005)

Epic 3: Delivery Scheduling & Tracking

Epic 4: Security & Access Control

Create epic

Backlog (9 work items)

Create a whiteboard to plan your work

SCRUM-1 Start here: Add your team's work

SCRUM-2 Connect your coding agent to Jira

SCRUM-3 Connect Atlassian Rovo MCP to your lo...

SCRUM-6 As a user u want to make payments

SCRUM-16 Story 3.1: Pause, Skip, or Reschedule ...

SCRUM-17 Story 3.2: Modify Delivery Within Cut...

SCRUM-18 Story 3.3: Track Delivery Fulfillment S...

SCRUM-19 Story 4.1: Restrict Access to Dietary D...

SCRUM-20 Story 4.2: Authenticated Access for D...

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Complete sprint

To Do 0

In Progress 5

In Review 0

Done 0

Create

Story 3.1: Pause, Skip, or Reschedule Delivery

SCRUM-16 5

Story 3.2: Modify Delivery Within Cutoff Window

Parent

Epic 2: Meal Plan Generation & Val...

SCRUM-17 3

Story 3.3: Track Delivery Fulfillment Status

SCRUM-18 3

Story 4.1: Restrict Access to Dietary Data

Parent

Epic 4: Security & Access Control

SCRUM-19 5

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3+

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Group

Complete sprint

To Do 0

In Progress 0

In Review 5

Done 0

Create

Create

Story 3.1: Pause, Skip, or Reschedule Delivery

SCRUM-16 5

Story 3.2: Modify Delivery Within Cutoff Window

Parent

Epic 2: Meal Plan Generation & Val...

SCRUM-17 3

Story 3.3: Track Delivery Fulfillment Status

SCRUM-18 3

Story 4.1: Restrict Access to Dietary Data

Parent

Epic 4: Security & Access Control

SCRUM-19 5

Create

The screenshot displays the Jira interface for a Kanban board. The top navigation bar includes a search bar, a '+ Create' button, and a 'See plans' button. The left sidebar contains navigation options: 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Recent', 'Recommended', 'Filters', 'Dashboards', 'Teams', 'Goals', 'Projects', and 'Customize sidebar'. The main content area shows the 'PESTUG24AM319' space with a Kanban board. The board has four columns: 'To Do' (0 items), 'In Progress' (0 items), 'In Review' (0 items), and 'Done' (5 items). The 'Done' column contains four story cards: 'Story 3.1: Pause, Skip, or Reschedule Delivery', 'Story 3.2: Modify Delivery Within Cutoff Window', 'Story 3.3: Track Delivery Fulfillment Status', and 'Story 4.1: Restrict Access to Dietary Data'. Each story card shows its parent epic, a progress bar, and a status icon. The bottom of the screen shows a 'Customize sidebar' button.



Complete SCRUM Sprint 2

This sprint contains 5 completed work items.
That's all of them - well done!

Cancel

Complete sprint

For you

Recent

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Try

Filters

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Filter

Epic

No epic

Epic 1: Dietary Profile Management (FR-002)

Epic 2: Meal Plan Generation & Validation (FR-001, FR-005)

Epic 3: Delivery Scheduling & Tracking

Epic 4: Security & Access Control

Create epic

Plan your sprint

Drag work items from the Backlog section or create new ones to plan the work for this sprint. Select Start sprint when you're ready.

Create

0 of 0 work items visible | Estimate: 0 of 0

Backlog (4 work items)

Create a whiteboard to plan your work Try

SCRUM-1

Start here: Add your team's work

To Do

Aug 30

-

SCRUM-2

Connect your coding agent to Jira

To Do

Sep 4

-

SCRUM-3

Connect Atlassian Rovo MCP to your local ...

To Do

Sep 9

-

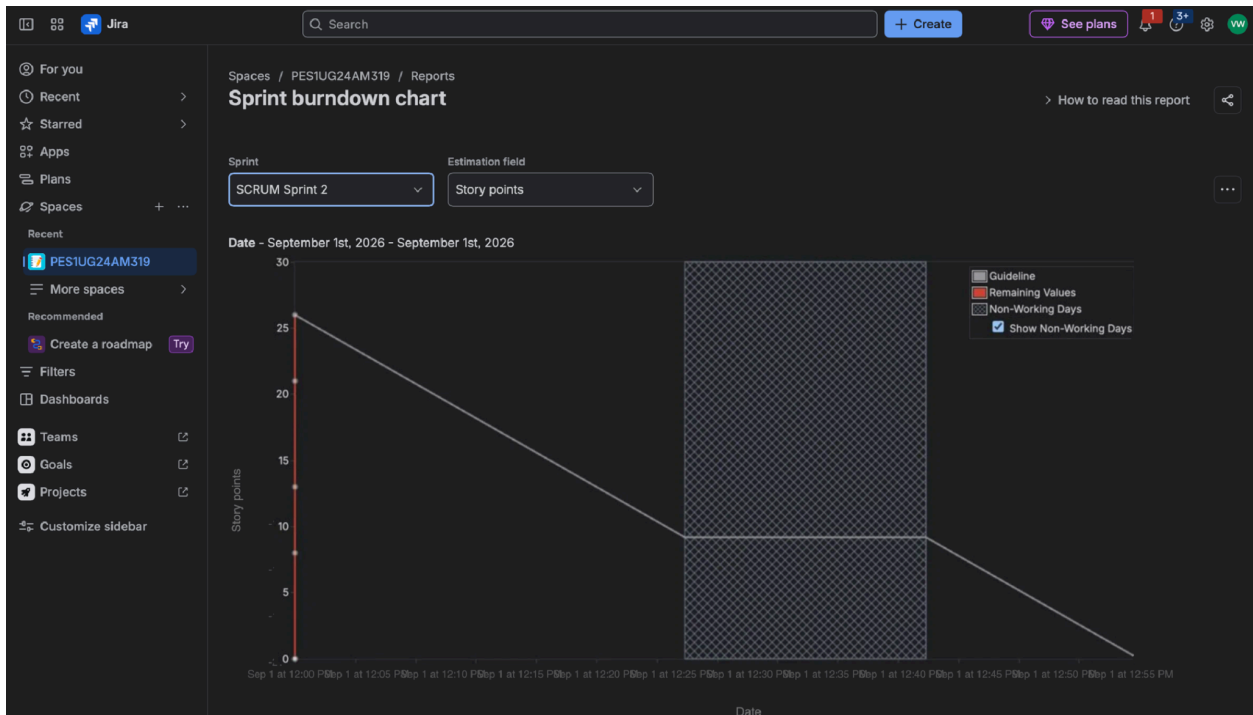
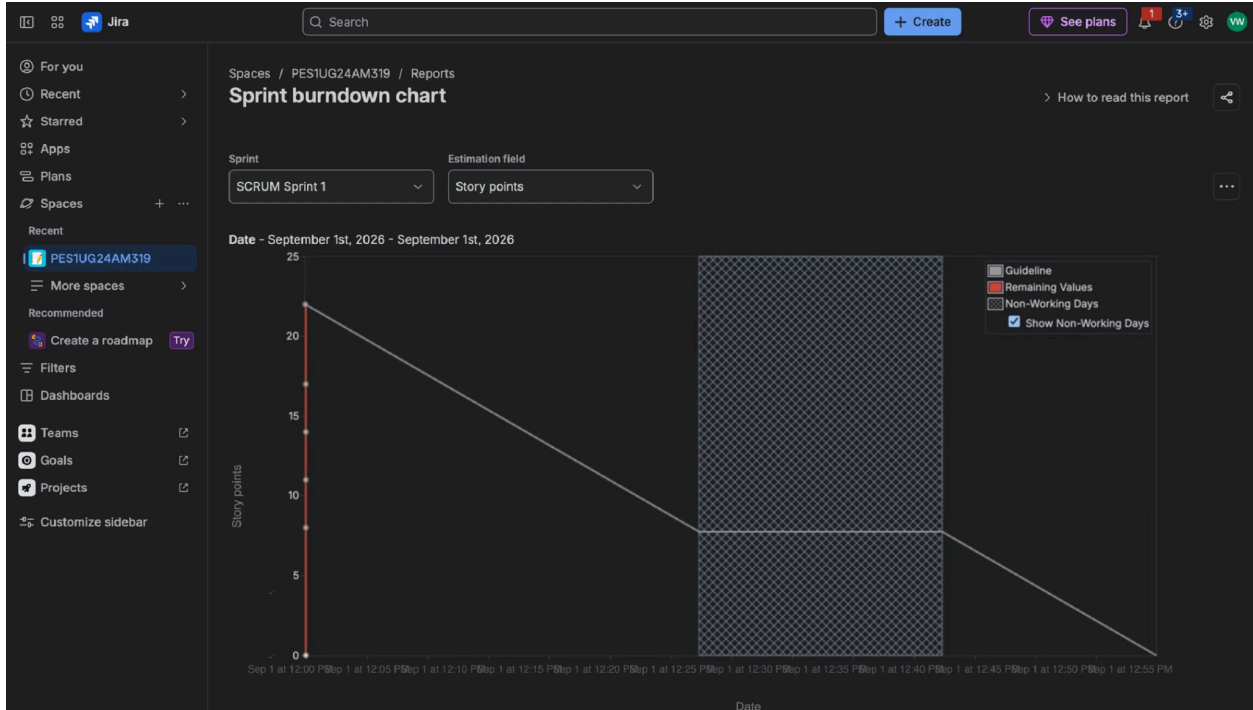
SCRUM-6

As a user u want to make payments

To Do

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Create



Reflection Questions:

Q1 Did your estimations reflect the actual effort?

A1 Yes, higher-complexity stories like meal plan generation and validation were given more story points (8 and 5) than simpler ones like profile updates (3). Since this was a simulated sprint rather than real development, actual effort couldn't be fully measured, but the relative point distribution matched our expectations of complexity.

Q2 Was your backlog well-prioritized?

A2 Yes, priority was set based on the FR/NFR priority levels from Lab 1. Core safety and functionality items (profile creation, meal plan generation and validation, data security) were marked High, while secondary features like dietitian review were Medium, ensuring critical work was tackled first.

Q3 How did your simulated sprint align with your plan?

A3 We split 10 stories evenly across two sprints (24 and 19 points), with Sprint 1 covering profile setup and meal plan generation, and Sprint 2 covering delivery management and access security. Both sprints were completed in full within our compressed 1-hour timebox.

Q4 What insights did the burndown chart give about your team's capacity?

A4 Both charts showed a near-vertical drop to zero rather than a gradual decline, since all stories were marked Done in quick succession during our short simulated sprint. In a real week-long sprint, this pattern would suggest work should be paced more evenly rather than completed in one burst near the end.